

Mohamed Ashour

AI Engineer · ML Engineer · LLM Engineer — RAG, MLOps, Data Engineering

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PROFESSIONAL SUMMARY

AI Engineer shipping production AI governance and compliance systems at SustainGRC, a UK-based AI-native GRC/ESG SaaS. Design and build AI modules under real production constraints — multi-tenant architecture, audit-grade data integrity, and EU AI Act alignment — across the full machine-learning lifecycle, from data pipelines to deployed LLM services on Azure.

Core strengths

- **LLM engineering** — RAG, agentic workflows, prompt engineering, vector search (pgvector), sentence-transformers, Groq Llama 3.3, Azure OpenAI (GPT-4o)
- **Full ML lifecycle** — model development, experimentation (MLflow, Optuna), and containerized deployment (Docker, GitHub Actions, Azure)
- **Production backends** — FastAPI, PostgreSQL, Redis, and scalable multi-tenant SaaS architecture (Next.js / Drizzle ORM)
- **Domain depth** — EU AI Act, GRC/ESG compliance, deterministic risk classification, and Shadow AI discovery

TECHNICAL SKILLS

- Core competencies** — LLM applications · RAG · Agentic AI · Prompt engineering · Embeddings and semantic search · Machine learning · MLOps · Data engineering · Time-series forecasting · Computer vision
- **Programming:** Python, TypeScript, JavaScript, SQL
 - **AI and LLMs:** RAG, Agentic AI, prompt engineering, Hugging Face Transformers, LangChain, pgvector, ChromaDB, embeddings, NLP
 - **Machine learning:** LightGBM, Scikit-Learn, PyTorch, Optuna, MLflow, time-series forecasting, fine-tuning
 - **Computer vision:** OpenCV, YOLOv5, MediaPipe, ResNet-50, CNNs
 - **Backend and data:** FastAPI, Next.js, Node.js, PostgreSQL, Redis, SQLAlchemy, Drizzle ORM, Pandas, NumPy
 - **Cloud and MLOps:** Azure ML, Azure OpenAI, Azure AI Foundry (Anthropic Claude), Docker, Docker Compose, GitHub Actions
 - **Domain:** GRC/ESG SaaS, multi-tenant architecture, EU AI Act, audit-grade AI systems

EXPERIENCE

AI Engineer — SustainGRC · Remote · 2026 – Present

AI-native GRC/ESG SaaS — end-to-end AI governance tooling

- Engineered a deterministic EU AI Act classification engine with five risk gates and a hash-chained audit trail, enabling audit-grade governance of AI systems across the platform
- Delivered AI-assisted compliance drafting powered by Anthropic Claude on Azure AI Foundry for structured rights-impact assessments
- Implemented AI discovery that matches network logs against a vendor signature library to surface unregistered AI tools
- Built a climate-risk data pipeline ingesting PostgreSQL tables with country-specific physical-risk multipliers
- Shaped the AI-governance product roadmap through product and competitor analysis

Data Engineer Intern — SustainGRC · Remote · 2025 – 2026

- Designed and shipped a Text-to-SQL feature translating natural-language questions into executable SQL via GPT-4o, with dual-layer security validation and 18/18 passing security tests
- Automated backup and disaster-recovery pipelines with integrity-validation checksums
- Applied FAIR data principles and automation to ESG data governance for sustainability and compliance reporting

Machine Learning Intern — CodeAlpha · Remote · 2024

- Fine-tuned a ResNet-50 image classification model for a computer-vision task
- Prototyped a music recommendation system using collaborative filtering
- Evaluated NLP and computer-vision models to inform development decisions

PROJECTS

RestAI — AI-Powered Restaurant Management SaaS

Python · FastAPI · Next.js · LightGBM · Groq Llama 3.3 70B · RAG · pgvector · Docker

- Architected a full-stack AI SaaS (FastAPI and Next.js) with JWT authentication and role-based access control
- Engineered LightGBM demand forecasting with Optuna tuning over lag and rolling-window features for time-series demand prediction
- Built a RAG assistant over menu and operational data using pgvector, sentence-transformers embeddings, and Groq Llama 3.3 70B
- Designed a hybrid recommender combining co-occurrence and content-based signals; 162 automated tests

Storefy — AI-First Multi-Tenant E-Commerce Platform

Next.js 15 · TypeScript · PostgreSQL · Drizzle ORM · Groq Llama 3.3 70B · Better Auth · Inngest · Paymob

- Engineered a multi-tenant platform with wildcard-subdomain routing and database-level data isolation per tenant
- Implemented a natural-language AI POS, natural-language analytics, and an AI business advisor
- Delivered generative storefront onboarding that produces a theme, layout blocks, and a starter catalog from a text description

Text-to-SQL Generator

Python · Azure OpenAI (GPT-4o) · Defense-in-depth validation · GitHub Actions

- Developed a production-hardened natural-language-to-SQL converter with multi-layer security validation and SQL-injection protection — 100% security test coverage (18/18 tests)

Semantic Book Recommender

Python · sentence-transformers · ChromaDB · LangChain · Gradio

- Developed a semantic-search recommendation system indexing 7,000+ books via vector embeddings, with roughly 67 ms average query time and zero API costs

Hand Gesture Recognition

Python · PyTorch · MediaPipe · MLflow

- Built a real-time recognition pipeline combining MediaPipe landmark extraction with a CNN, tracking experiments in MLflow

EDUCATION

B.Sc. Computer Science — Zagazig University, Egypt · 2023

Coursework: algorithms, database systems, distributed computing

CERTIFICATIONS AND PROGRAMS

- **Supervised Machine Learning** — DeepLearning.AI
- **AWS Machine Learning Foundations** — Amazon
- **Claude Code in Action** — Anthropic
- **Microsoft Machine Learning Track** — Digital Egypt Pioneers Initiative (DEPI)
- **AI Diploma (150 hours)** — Array Institute: MLOps, deep learning, data engineering
- **IEEE Delta Academy** — AI Track (Member)